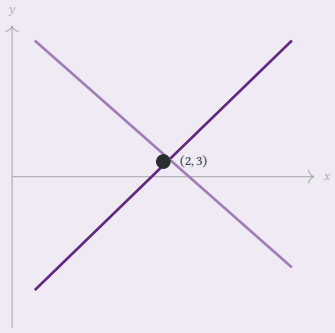
QOIDALAR VARAG'I – 1/2

QOIDA · Sistema yechimi – bu kesishish nuqtasi

Yechim – ikkala tenglamani ham to'g'ri qiladigan (x, y) juftligi. Grafik jihatdan bu ikki chiziqning kesishgan nuqtasi.



Tekshirish: $(2, 3)$ sistemaning yechimimi?
$x + y = 5$: $2 + 3 = 5$ – to'g'ri.
$2x - y = 1$: $2(2) - 3 = 1$ – to'g'ri.
Ikkalasi ham to'g'ri – demak, yechim.

QOIDA · O'rniga qo'yish usuli (substitution)

Bitta tenglamada o'zgaruvchi ikkinchisi orqali ifodalangan bo'lsa, shuni boshqa tenglamaga qo'yamiz.

$y = 2x - 1, \quad 3x + y = 9$ ← y ifodalangan

↓

$3x + (2x - 1) = 9$ ← o'rniga qo'yamiz

↓

$5x - 1 = 9 \Rightarrow x = 2$ ← yechamiz

↓

$y = 2(2) - 1 = 3 \Rightarrow (x, y) = (2, 3)$

QOIDA · Qo'shish-ayirish usuli (elimination)

Mos koeffitsientlar bir-birini yo'qotadigan tarzda ko'paytirib, so'ng qo'shamiz yoki ayiramiz.

$x + 2y = 11, \quad 3x + y = 18$ ← y ni yo'qotamiz

↓

$3x + y = 18 \ (\times 2) \Rightarrow 6x + 2y = 36$ ← 2-tenglamani ×2

↓

$(6x + 2y) - (x + 2y) = 36 - 11 \Rightarrow 5x = 25$ ← ayiramiz

↓

$x = 5 \Rightarrow y = 3 \Rightarrow (x, y) = (5, 3)$

QOIDALAR VARAG'I – 2/2

DIQQAT · Ayirishda HAR BIR had ishorasini o'zgartiring

$(6x+2y)-(x+2y)$ ni hisoblaganda, ikkinchi qavsdagi **barcha** hadlarning ishorasi o'zgaradi – faqat birinchisi emas:

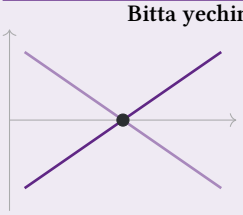
$6x + 2y - x - 2y = 5x + 0y = 5x$ **($2y$ ni unutish y ni yo'qotmay qo'yadi!)**

QOIDA · Maxsus holatlar: yechimsiz va cheksiz ko'p yechim

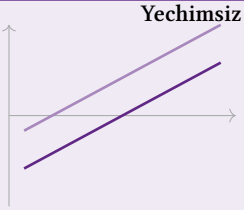
x va y koeffitsientlari nisbatini solishtiring:

Nisbatlar	Natija
Koeffitsientlar nisbati teng emas	Bitta yechim
Koeffitsientlar teng, konstanta boshqacha	Yechimsiz (parallel chiziqlar)
Koeffitsientlar va konstanta hammasi teng	Cheksiz ko'p yechim (bir xil chiziq)

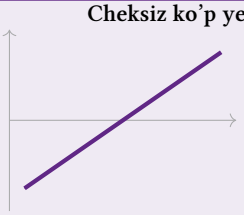
Bitta yechim



Yechimsiz



Cheksiz ko'p yechim



Protsedura ichidan belgi: o'zgaruvchi yo'qotilayotganda **ikkalasi ham yo'qolib**, noto'g'ri tenglik chiqsa ($0 = 12$) – **yechimsiz**; har doim to'g'ri tenglik chiqsa ($0 = 0$) – **cheksiz ko'p yechim**.

DIQQAT · Doim tekshiring: nima so'ralmoqda?

$x + y = 10$, $x - y = 2$. **Savol:** xy ning qiymati nechaga teng? (x yoki y emas!)

Yechamiz: $x = 6$, $y = 4$. Savol xy ni so'ragan, shuning uchun oxirgi qadam – ko'paytirish: $6 \times 4 = 24$.

Javob: $xy = 24$ – $x = 6$ yoki $y = 4$ emas.